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Utility of VOMS, SCAT3, and ImPACT Baseline Evaluations for Acute Concussion Identification in Collegiate Athletes

Findings From the NCAA-DoD Concussion Assessment, Research and Education (CARE) Consortium

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Background: The Vestibular/Ocular-Motor Screening (VOMS) is a valuable component of acute (<72 hours) sports-related concussion (SRC) assessments and is increasingly used with the Immediate Post-concussion Assessment and Cognitive Testing (ImPACT) instrument and the third edition of the Sport Concussion Assessment Tool (SCAT3). Research has suggested that VOMS acute postinjury scores are useful in identifying acute concussion. However, the utility of preseason baseline measurements to improve diagnostic accuracy remains ambiguous. To this end, there is a need to determine how reliable VOMS baseline assessments are across years and whether incorporating individuals' baseline performance improves diagnostic yield for acute concussions.

Purpose: To analyze VOMS, SCAT3, and ImPACT to evaluate the test-retest reliability of consecutive-year preseason baseline assessments to directly compare the diagnostic utility of these tools when incorporating baseline assessments versus using postinjury data alone to identify acute SRC.

Study Design: Cohort study (diagnosis); Level of evidence, 2.

Methods: Preseason and postinjury VOMS, SCAT3, ImPACT Post-Concussion Symptom Scale (PCSS), and ImPACT composite scores were analyzed for 3958 preseason (47.7% female) and 496 acute (≤48 hours) SRC (37.5% female) collegiate athlete evaluations in the National Collegiate Athletic Association—Department of Defense Concussion Assessment Research and Education Consortium. Descriptive statistics, Kolmogorov-Smirnov significance, and Cohen *d* effect size were calculated. Consecutive-year baseline reliability was evaluated for a subset of 447 athlete encounters using Pearson *r*, Cohen κ, Cohen *d*, and 2-way mixed intraclass correlation coefficients (ICCs). Wilcoxon signed rank tests were used to determine the statistical significance between population performances, and the 90% reliable change index (RCI) was calculated from the test-retest results. Preseason to postinjury change scores were then calculated from each tool's RCI. Finally, receiver operating characteristic (ROC) curve analyses were conducted, and DeLong method was used to compare the area under the curve (AUC) of raw postinjury scores versus change scores from preseason baseline assessments. Potential effects of sex, medical history (learning disorders or attention-deficit/hyperactivity disorder), and outlier data were also explored.

Results: Effect sizes were large, and overall predictive utilities were clinically useful for postinjury VOMS Total (d = 2.44; AUC = 0.85), the SCAT3 Symptom Evaluation total severity score (d = 1.74; AUC = 0.82), and the ImPACT PCSS total severity score (d = 1.67; AUC = 0.80). Comparatively, effect sizes were small and predictive utilities were poor for Standardized Assessment of Concussion (SAC), modified Balance Error Scoring System (mBESS), and all ImPACT composites (d = 0.11-0.46; AUC = 0.48-0.59). Preseason baseline test-retest reliability was poor to moderate (r = 0.23-0.52; κ = 0.32-0.36; ICC = 0.36-0.68) for all assessments

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except ImPACT Visual Motion Sensitivity (r = 0.73; ICC = 0.85). Incorporating baseline scores for VOMS Total, SCAT3 (Symptom Evaluation, SAC, mBESS), ImPACT PCSS, or ImPACT composites did not significantly improve AUCs.

Conclusion: VOMS Total and symptom severity (SCAT3, PCSS) total scores had large effect sizes and clinically useful AUCs for identifying acute concussion. However, all tools demonstrated high within-patient test-retest variability, resulting in poor reliability. The findings in this sample of collegiate athletes suggest that incorporating baseline assessments does not significantly increase diagnostic yield for acute concussion.

Keywords: sport-related concussion; VOMS; SCAT; ImPACT; Reliable Change Index

Sports-related concussions (SRCs) are common in contact sports and can be challenging for health care providers to evaluate and manage.[20,36] Given a lack of laboratory biomarkers and insufficient sensitivity of diagnostic imaging, clinicians assessing SRC commonly rely on a relatively standardized multidomain battery of assessments covering somatic, cognitive, affect, vestibular, and visual-oculomotor symptoms.[9,22,28,36] The Vestibular/Ocular-Motor Screening (VOMS) instrument is a validated[16,29] assessment of the vestibular and oculomotor systems with high diagnostic accuracy for acute concussion in both adolescent and collegiate athlete populations.[38] It is increasingly being used alongside other tools such as the Sport Concussion Assessment Tool (SCAT) and the Immediate Post-concussion Assessment and Cognitive Testing (ImPACT) tool for concussion assessment and management, including during preseason baseline testing.[36]

These tools have been shown to improve the accuracy of concussion identification in acute evaluations,[11,18,41] and they are commonly conducted in mandatory preseason concussion evaluation settings and at the acute phase of injury.[36] However, controversy has arisen over the comparative diagnostic value of using raw postinjury scores alone versus considering change scores based on the difference between the postinjury performance and the most recent preseason baseline test. Some studies have shown improved SRC detection with change scores,[19,51] whereas others have found no improvement in diagnostic utility compared with raw scores.[11,47] One type of change score used to provide reference ranges for athlete performance is the Reliable Change Index (RCI).[26] Athletes whose changes between preseason baseline and acute SRC scores fall within their RCI may have their performance considered within normal ranges, whereas those whose changes fall outside their RCI are considered to have abnormal performance. RCIs are easy to use and appealing to clinicians when speed of diagnosis is critical (eg, sideline, battlefield). Given this, interpreting a patient's performance using the RCI has been proposed for the SCAT[8,12] and is commonly used when analyzing ImPACT performance changes.[25] However, conflicting evidence regarding the efficacy of RCI-derived change scores calls into question their overall clinical utility compared with referencing postinjury scores alone. The use of preseason change (PS Change) scores for SCAT and ImPACT via the RCI has shown mixed benefit compared with using raw postinjury scores.[7,25,47] The symptom component of VOMS is similar to SCAT and the Post-Concussion Symptom Scale (PCSS) of ImPACT. As such, there is a strong need to directly compare the diagnostic utility between raw and RCI change score methods for the VOMS so clinicians can be better informed when making diagnostic decisions.

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Overall, neither the use of RCI for accurately detecting changes between preseason and acute SRC performance nor their variance across unique subpopulations has been explored fully. Whereas general change score explorations have been conducted, literature is lacking on the possibility of sex or premorbid medical history effects such as migraine headaches, the number of previous concussions, learning disorders (LD), or attention-deficit/hyperactivity disorder (ADHD). This is despite known differences in these populations, all of which have been shown to demonstrate neurocognitive, balance, and symptom score performance differences.[49,53] Similarly, there is limited literature exploring the effect that potential outlier data points have on change scores and diagnostic utility of evaluations.

The RCI is directly calculated from performance variance over time and uses both the SD of a test's scores and the correlation between test and retest scores in its derivation.[26] RCIs are ideal for clinical use because they help clinicians determine what level of performance change should be expected within the same individual. RCIs assume that the results of a tool are Gaussian in nature and vary smoothly about the mean (eg, are not heteroscedastic).[26] Published reliability metrics for acute concussion assessment tools including the VOMS, SCAT, and ImPACT suggest relatively high inherent variance and less than ideal reliability from year to year.[7,47] This suggests that change scores may not be beneficial for clinicians assessing acute concussion.[7,11,41,47]

Concussion assessments should be consistent across testing times for healthy athletes (eg, at consecutive preseason baseline assessments). If a tool has high within-patient variance across healthy time points, baseline performance may not serve as a reliable reference for future comparison.[47] Varying methods of assessing reliability for common concussion assessments, including Pearson r and intraclass correlation coefficient (ICC), have been used.[7,11,40,41,44] Because many symptom-based assessments are known to display significant skew, the Cohen κ has been proposed for use in place of ICC.[7] However, test-retest variance alone does not fully convey the predictive utility of a tool. To provide a broader view of a tool's overall predictive power, receiver operating characteristic (ROC) curves and their area under the curve (AUC) can be evaluated.[33] ROC curves are especially useful when evaluating a new diagnostic test because they allow comparisons between multiple tests, even when these tools use different data.[32] The use of ROCs to evaluate concussion assessment utility is increasing, with results aiding clinicians' understanding of the diagnostic value that various tools offer.[11,18,41]

Clinicians understandably face significant pressure to maximize accurate and timely concussion detection. Yet, conducting preseason baseline evaluation assessments on every athlete places significant resource burdens, in terms of time and costs, on both athletes and organizations.[44] Given the lack of consensus around the value of baseline assessments, leading sports organizations have begun to suggest that this testing, although potentially useful, may not be essential for appropriate SRC management.[22,36] Definitive research, using a large sample and thorough analytical techniques, would assist organizations in making the most appropriate decision for their athletes. Specifically, research is needed to evaluate the test-retest reliability of preseason baseline assessments for the VOMS relative to ImPACT and the third edition of SCAT (SCAT3) to directly compare the diagnostic utility of these tools when incorporating baseline assessments versus using normative (raw) data for identifying acute SRC.

Using the Concussion Assessment Research and Education (CARE) Consortium data set, this study evaluated consecutive-year baseline test-retest reliability for components of VOMS, SCAT3, and ImPACT. New RCIs were calculated for each assessment and used to generate individualized PS Change scores from baseline performance to compare the diagnostic accuracy of these scores versus postinjury-only performance. Additional supplementary analyses explored possible effects of sex, premorbid medical history, and outlier data points.

METHODS

The complete methods on participant recruitment, enrollment, and assessment for the CARE Consortium have been previously published.[7] In summary, National Collegiate Athletic Association (NCAA) varsity-level student athletes and military service academy cadets were invited to participate and enrolled in the study after providing informed consent. Institutional review board approval was obtained from individual institutional review boards and the US Army Human Research Protection Office. All study activities were completed in accordance with the Declaration of Helsinki. Whereas athletes included in this analysis were required to have at least 1 complete VOMS and SCAT3 Symptom Evaluation assessment, they were not excluded if components of ImPACT were missing. However, per the ImPACT manual, ImPACT assessments with Impulse Control scores >30 were excluded (n = 3) for poor reliability.[25] Missing scores for the Standardized Assessment of Concussion (SAC), the modified Balance Error Scoring System (mBESS), or any of the ImPACT components were considered not-a-number and did not affect overall statistics.

Participants

Of the 34,709 unique athletes in the CARE data set as of February 23, 2021, a total of 3464 individuals from 6 NCAA Division I, 2 Division II, and 2 Division III institutions completed VOMS, SCAT3 (Symptom Evaluation, SAC, mBESS), and ImPACT assessments (PCSS and composite scores) and were screened for inclusion in the current investigation. Assessments were conducted either at the athlete's initial preseason evaluation or at any acute ($\leq$48 hours) concussion time point. Participants with a history of vestibular disorders, moderate or severe brain injury, and/or brain surgery were excluded (n = 20), resulting in a final cohort of 3444 participants and 3958 preseason evaluations. Among the 3444 athletes, 388 had data that were previously analyzed by this research group.[18]

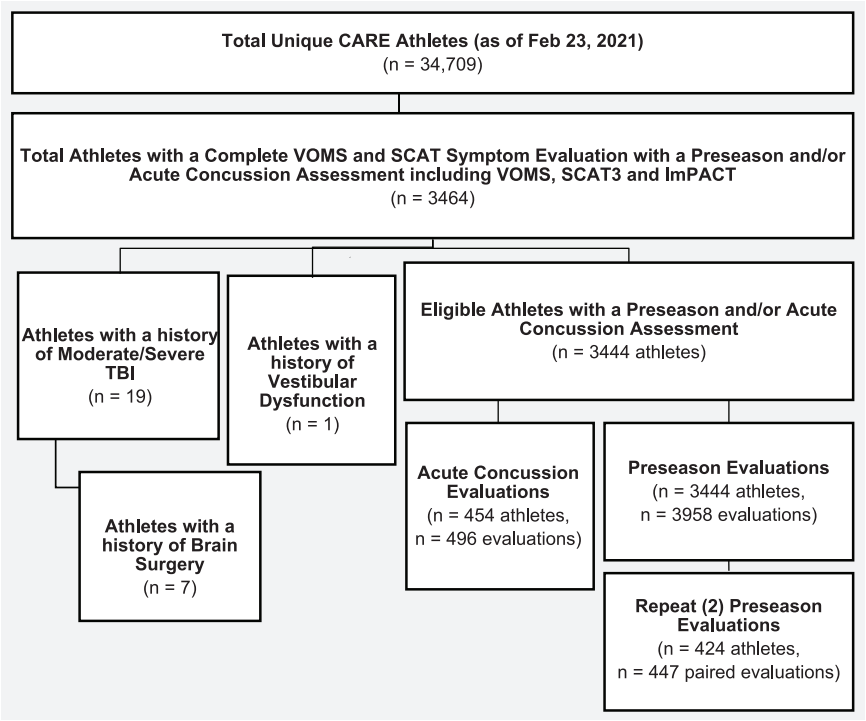

Figure 1. Athlete evaluation selection schematics. CARE, Concussion Assessment Research and Education; ImPACT, Immediate Post-concussion Assessment and Cognitive Testing; SCAT, Sport Concussion Assessment Tool; TBI, traumatic brain injury; VOMS, Vestibular/Ocular-Motor Screening.

A total of 454 athletes had sustained at least 1 acute concussion, defined as "a change in brain function after a force to the head, which may be accompanied by temporary loss of consciousness, but is identified in awake individuals with measures of neurologic and cognitive dysfunction."[6,10] Because 38 participants experienced 2 concussions and 2 participants experienced 3 concussions, the data set included a total of 496 acute concussion evaluations (Figure 1).

For the test-retest evaluations, a total of 424 athletes with consecutive-year preseason baseline evaluations were identified. Athletes participated in activities at 2 NCAA Division I institutions, 1 Division II institution, and 1 Division III institution. To reflect typical consecutive-year test-retest conditions seen at some athletic institutions, repeat preseason evaluations were included if they were conducted $\geq$9 months (273.75 days) and $\leq$15 months (456.25 days) apart. Within this population, 80 athletes (18.9%) sustained a concussion between repeat baseline evaluations, but they did not perform significantly differently from the total subset and were retained in the analysis. Among the 424 athletes with at least 2 completed evaluations, 21 athletes had completed 3 consecutive evaluations and 1 athlete had completed 4 consecutive evaluations, for a total of 447 test-retest evaluations.

Assessment Variables

Vestibular/Ocular-Motor Screening. The VOMS tool is a free, quick, pen-and-paper assessment of vestibular and oculomotor symptoms using 8 component tasks.[38] Tasks include (1) smooth pursuits (horizontal and vertical), (2) horizontal saccades, (3) vertical saccades, (4) near point of convergence (NPC) symptoms, (5) NPC distance, (6) horizontal vestibulo-ocular reflex, (7) vertical vestibulo-ocular reflex, and (8) visual motion sensitivity. Before VOMS administration, patients are asked to report pretest (baseline) symptoms on an 11-point Likert scale (0-10) across 4 symptom categories: headache, nausea, fogginess, and dizziness. Symptoms are then reassessed at the conclusion of each task to determine whether the vestibular-oculomotor activity exacerbated symptoms. These symptoms are often summed individually, providing a symptom score for each task (maximum, 40 per task),[38] but can also be summed across all tasks, providing an overall total symptom score for the 7 components (maximum, 280). In addition to the symptom report, VOMS NPC distance (in cm) is measured and averaged across 3 trials, resulting in a mean NPC distance. This results in 2 different scoring systems for VOMS: symptom severity (either component task or overall total) and mean NPC distance. Both VOMS Total symptom severity score and mean NPC distance were evaluated in this study.

Sport Concussion Assessment Tool, Third Edition. The SCAT3 is a multidimensional assessment of physical, neurological, symptomatic, cognitive, and spinovestibular systems.[21] Although SCAT is currently in its 5th edition, the SCAT3 was the most current edition at the start of the CARE Consortium, and elements of it were selected for

use throughout the duration of the CARE study, including the full BESS, the SAC, and a graded symptom evaluation, summarized below.

The mBESS, a subset of the BESS, assesses the spinovestibular system, with participants asked to conduct 3 single-leg and tandem-leg stances barefoot (or with socks) on a firm surface and with eyes closed for 20 seconds each.[21] Participants begin at zero and receive a positive score for each balance deviation, up to a total of 10 deviations per stance (maximum, 30). Although CARE protocols included both modified and full BESS (mBESS followed by a repeat of all 3 stances on an AIREX® foam pad),[6] the standard SCAT3 includes only mBESS. As such, only mBESS scores were evaluated in this study.

The SAC assesses neurocognitive parameters including orientation, immediate and delayed recall for a 5-word list, and concentration evaluations.[21] Although the 10-word list for SAC immediate and delayed recall was introduced in year 3 of CARE, only the 5-word list was used for data collection to enable data analysis across the lifecycle of the study. Scores can be calculated either by component or as the sum of all components. The SAC is reverse scored, with a perfect performance of 30 points and points taken away for errors. In the analyses conducted for this study, descriptive statistics for SAC were calculated based on the standard reverse-scoring method, and AUC analyses (see below) were calculated using an inverted and absolute SAC score (eg, where scores began at zero and increased with each error to a maximum of 30).

The final component of SCAT3 assessed in this study was the SCAT3 Symptom Evaluation, a 22-item neurological and somatic questionnaire on a 7-point Likert scale (0-6). In the most recent Concussion in Sport Group consensus statement, the SCAT3 was updated to the SCAT5, but no changes were made to the items evaluated.[15] Items include common symptoms such as headache, nausea, fogginess, and dizziness, as well as balance problems, sensitivity to light, and blurred vision. The Symptom Evaluation questionnaire can be scored as either the total number of items with a positive symptom report (maximum, 22) or the sum of the severity of all symptoms (maximum, 132). For this study, only the evaluations' total symptom severity score was evaluated.

Immediate Post-concussion Assessment and Cognitive Testing (ImPACT). The ImPACT tool is the most commonly used computerized neurocognitive assessment tool for assessing cognitive changes secondary to concussion.[1] It consists of a demographic information section, a postconcussion symptom scale, and multiple cognitive tasks that are aggregated into 4 composite scores: (1) Verbal Memory, (2) Visual Memory, (3) Visual Motor Speed (VMS), and (4) Reaction Time (RT). High scores on Verbal and Visual Memory and processing speed components indicate better performance, whereas lower scores on RT indicate better performance. Because higher Visual Memory and Verbal Memory component scores indicate better performance, raw scores were used to calculate descriptive statistics, whereas inverted and absolute scores (where maximum performance begins at zero and degrades until reaching a minimum performance score of 100) were used to calculate AUC curves (see below).

The final component of ImPACT is the PCSS. The ImPACT PCSS is a 22-item self-report inventory designed to evaluate concussion-related symptoms on a scale of 0 (none) to 6 (severe). Items reflect physical-somatic (eg, headache, dizziness), cognitive (eg, difficulty remembering, difficulty concentrating), emotional (eg, anxiety, depressed mood), and sleep-related symptom categories. For this study, the total PCSS symptom severity score, which ranges from 0 to 132, was evaluated.

Data Analysis

Data analysis to examine relationships between preseason and acute postinjury performance within the components of VOMS, SCAT3, and ImPACT was conducted using custom scripts in R (R Core Team) over 4 distinct phases.[46,54] First, general descriptive statistics assessing raw preseason and acute postinjury scores were run on all components. Second, consecutive-year baseline test-retest reliability was assessed using multiple analytic tests, including calculating the RCI for each component. Third, PS Change scores were calculated using each component's RCI. Fourth, these were then compared against raw scores in an ROC curve analysis to determine the predictive utility of each method.

General Statistical Approaches and Descriptive Statistics. Standard descriptive statistics included the mean and variance between time points for each tool. For demographic discrimination, an independent-samples means test was used to calculate statistical significance between male and female participants. The Kolmogorov-Smirnov test was used to calculate statistical significance between concussed and nonconcussed populations, whereas unpaired Wilcoxon signed rank tests were used to calculate significance between test-retest performance outcomes and between the outcomes in male, female, and LD/ADHD populations. Effect sizes (eg, the magnitude of difference between 2 distributions) were calculated using Cohen d, where $d \leq 0.40$ was considered minimal; $0.40 < d \leq 0.80$, moderate; and $d > 0.80$, large.[3,23] For all analyses, we used an a priori type I error rate of $P < .01$.

Test-Retest Reliability. Test-retest reliability was assessed using the subset of 424 participants who had consecutive-year preseason baseline evaluations. Multiple test-retest measures were used because all have been used to assess VOMS, SCAT, or ImPACT within the concussion literature.[7,11,25,44] Assessment measures included a 2-way random average score intraclass correlation (ICC2k, hereby referred to as ICC), Cohen d, Cohen κ, Pearson r, and the 90th CI RCI (90th RCI). Correlations were considered poor if $\leq$0.40, moderate if $>$0.40 and $\leq$0.70, and good if $>$0.70.[30] Kappa has been proposed as ideal for use with the skewed data seen in many symptom-based concussion assessments and has previously been used to describe test-retest performance of VOMS, the SCAT3 Symptom Evaluation, and ImPACT PCSS

within the CARE data set.[7] It was included for all symptom-based assessment tools in this analysis to enable comparison with other literature. Before we estimated κ, the VOMS Total score was categorized as 0, 1, 2, and ≥3, whereas SCAT3 Symptom Evaluation and ImPACT PCSS total severity scores were categorized as 0, 1, 2, 3, and ≥4.[7] All other correlations were calculated using the raw scores of each respective tool.

All RCIs were calculated using the Iverson method[25] adapted from Jacobson and Truax,[26] summarized in Formula 1:

Formula 1: RCI Derivation Formula

$$SEM_1 = SD_1 \text{ x } \sqrt{(1-r_{12})}$$
$$SEM_2 = SD_2 \text{ x } \sqrt{(1-r_{12})}$$
$$RCI = 1.645 \text{ x } \sqrt{(SEM_1{}^2 + SEM_2{}^2)},$$

where SEM = standard error of measurement at initial (1) or consecutive (2) evaluations, SD = standard deviation at initial (1) or consecutive (2) evaluations, r_{12} = Pearson correlation coefficient between initial (1) and consecutive (2) evaluations, and RCI = reliable change index for 90th CI.

Diagnostic Utility of PS Change Versus Raw Scores. After the 90th RCIs were determined for each tool, changes in performance between consecutive evaluations were used to determine whether the difference between an individual's past performance was within or outside the range of reliable change for each tool, using Formula 2:

Formula 2: PS Change With RCI

$$\Delta T_{non-Concussed} = T_2 - T_1$$
$$\Delta T_{Concussed} = T_{acute\ injury} - T_1$$
$$PS\ Change = |RCI - \Delta T|,$$

where ΔT = the difference between consecutive assessment tool (T) scores, T_1 = initial preseason assessment tool score, T_2 = consecutive preseason baseline tool score (if present), and $T_{acute\ injury}$ = assessment tool score at the acute concussion time point.

To determine whether preseason baseline evaluations were beneficial for acute concussion identification, we compared PS Change score results with raw postinjury performance via the Youden method of ROC analyses[4] using evaluations within the test-retest data set (n = 447) and the total cohort, which contained data on evaluations of athletes who underwent at least 1 preseason evaluation (n = 3958) and then experienced a concussion (n = 496). The Youden method was chosen because it represents the best cutoff value for an assessment tool when no preference for maximizing sensitivity or specificity exists, such as seen in these acute concussion assessments. The ROC analyses were conducted to compare predictive power and determined the AUC, sensitivity, and specificity of each assessment tool across the 2 methods: (1) Raw, using only the raw scores for each respective evaluation time point, and (2) PS Change, using the individualized PS Change scores between each evaluation time point.[17,50] AUC values range from zero to 1. An AUC falling along the chance line (0.50) indicates the test has a 50% probability of correctly identifying the desired condition, equating to a 100% guess rate for the test (very poor). An AUC >0.80 indicates an 80% probability of correct identification (equivalent to a 40% guess rate) and is considered acceptable and clinically useful.[32] Of note, AUCs <0.50 indicate that the test is providing information inverse to the diagnostic question. That is, for that test, an improper decision rule was used.[33] The DeLong method of AUC comparisons was used to determine whether differences in AUC results were statistically significant.[13] Figure 2 summarizes the analysis strategy to evaluate the reliability and predictive power of Raw versus PS Change scoring methods for VOMS, SCAT3, and ImPACT.

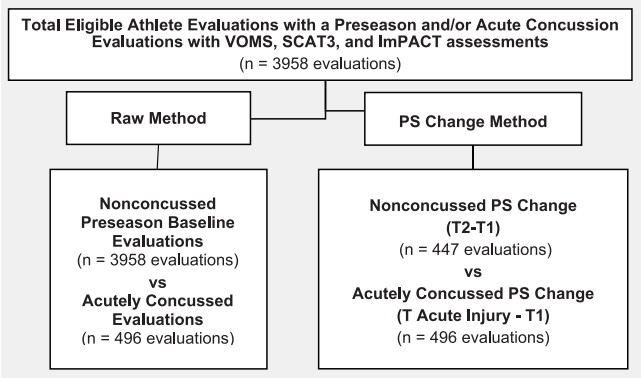

Figure 2. Analysis strategy for diagnostic utility of preseason change (PS Change) versus raw scores. ImPACT, Immediate Post-concussion Assessment and Cognitive Testing; SCAT, Sport Concussion Assessment Tool; VOMS, Vestibular/Ocular-Motor Screening; T, Time/instance of evaluation.

RESULTS

Descriptive Statistics and Effect Sizes

A total of 3958 athlete evaluations (1889 female; 2069 male; age, 19.44 ± 1.48 years) were included in the study (Table 1). Among the 496 total unique concussions, 328 (66.1%) entailed vision or balance problems (as noted via a SCAT3 Symptom Evaluation total severity score >0 for sensitivity to light, blurred vision, dizziness, or balance problems) at the time of the acute evaluation. A total of 3 athletes (0.60%) had positive computed tomography findings for facial fractures at the time of injury, but they had no other structural or positive magnetic resonance imaging or neurological findings and were retained in the analysis. The overall median number of days before return to play was 13.20 days. Individual sample test analyses revealed significant differences of age (older male participants), history of concussions (more in male participants), and acute SRCs sustained (more in male participants). All other differences between sexes were not significant ($P > .01$).

Figure 3 shows histogram distributions and summarizes the descriptive statistics for VOMS, SCAT3, and

TABLE 1
Population Characteristics[a]

	Female	Male	Total	*P* Value
Preseason characteristics				
Total preseason encounters	1889 (47.7)	2069 (52.3)	3958 (100)	<.0001
Age (y)	19.18 ± 1.32	19.61 ± 1.56	19.44 ± 1.48	<.0001
History of attention-deficit/hyperactivity disorder	128 (6.5)	166 (7.5)	276 (7.0)	.2410
History of learning disorder	54 (2.7)	72 (3.3)	113 (2.9)	.3913
History of migraine headaches	143 (7.3)	148 (6.7)	267 (6.8)	.4922
Previous concussions at baseline	0.35 ± 0.69	0.44 ± 0.77	0.37 ± 0.72	
0 concussions	1240 (65.64)	1239 (59.88)	2479 (62.63)	<.0001
1 concussion	299 (15.83)	374 (18.08)	673 (17.00)	
2 concussions	66 (3.49)	111 (5.36)	177 (4.47)	
3 concussions	21 (1.11)	26 (1.26)	47 (1.19)	
4 concussions	3 (0.16)	8 (0.39)	11 (0.28)	
5 concussions	1 (0.05)	2 (0.10)	3 (0.08)	
6 concussions	2 (0.12)	4 (0.19)	6 (0.15)	
Not Reported	18 (0.95)	30 (1.45)	48 (1.21)	
Acute sport-related concussion characteristics				
Acute concussions	186 (37.5)	310 (62.50)	496 (100)	<.0001
Loss of consciousness	6 (3.50)	14 (5.50)	20 (4.03)	.4597
Posttraumatic amnesia	7 (4.10)	23 (9.00)	30 (6.05)	.0768
Retrograde amnesia	5 (2.90)	12 (4.70)	17 (3.43)	.4964
Vision symptoms (sensitivity to light or blurred vision)	127 (68.30)	158 (51.00)	285 (57.50)	.0305
Balance symptoms (dizziness or balance problems)	110 (59.10)	136 (43.90)	246 (49.60)	.03934
Median days to return to play (mean ± SD)	13.0 (19.55 ± 22.45)	13.90 (26.85 ± 45.30)	13.20 (23.97 ± 38.09)	.0717

[a]Data are expressed as n (%) or mean ± SD unless indicated otherwise.

ImPACT component scores. VOMS Total (d = 2.44), SCAT3 Symptom Evaluation (d = 1.74), and ImPACT PCSS (d = 1.67) demonstrated large effect sizes and significant increases after a concussion ($P < .01$). Conversely, mBESS, SAC, mean NPC, ImPACT Visual Memory, ImPACT VMS, and ImPACT RT displayed significant differences between preseason and acutely concussed time points but had only moderate to small clinical significance ($d \leq 0.46$). Changes for ImPACT Verbal Memory were not significant ($P > .01$). Due to the nonparametric distribution of most assessments, large differences were observed between the mean and SD and the median and 90% CI at both preseason and acute SRC time points.

Subanalyses of sex-specific data sets revealed differences in descriptive statistics and effect sizes at both the preseason and the acute SRC time points (Appendix Figures A1-A6, available in the online version of this article). Female participants had higher symptom scores at both time points and larger differences in performance between time points for all other assessments. These differences led to larger effect sizes than the male-only data and were significant ($P < .01$) at the preseason evaluation for all assessments except mean NPC distance. Conversely, at the acute SRC time point, these differences were significant ($P < .01$) for VOMS, the SCAT3 Symptom Evaluation, and the PCSS. Similar analyses of participants with a history of migraines (Appendix Figure A10, available online) and those with a history of ≥3 concussions (Appendix Figure A16, available online) revealed a higher symptom burden before and after injury for VOMS Total, SCAT3 Symptom Evaluation, and PCSS and a shift in ImPACT Visual Memory performance ($P < .01$ for all). Athletes with a medical history of LD/ADHD (Appendix Figure A22, available online) demonstrated no significant difference from those without a history of LD/ADHD (Appendix Figure A19, available online). However, athletes with LD/ADHD did demonstrate a performance similar to the female athletes data, noting higher preseason and acute SRC symptom severity burdens and demonstrating larger differences between preseason and acute SRC time points for all other assessments.

Test-Retest Reliability

Preseason test-retest analyses on 424 athletes (n = 447 evaluation pairs) revealed relatively high interpatient variability for most VOMS, SCAT3, and ImPACT components (Figure 4). Modified Brinley plots[5] showed correlations between consecutive preseason evaluations. At the group level, test-retest differences displayed an overall uniform distribution between years, with minimal effect sizes for any tool ($d \leq 0.17$), except for the 5-word SAC (d = 0.57). This equated to the distribution of performance being similar across years for each tool and suggested minimal practice effects. Traditional test-retest analysis methods revealed high within-patient variability for all tools, with VOMS Total (r = 0.23; ICC, 0.36; κ = 0.32), SCAT3 Symptom Evaluation (r = 0.36; ICC, 0.53; κ = 0.36), mBESS (r = 0.52; ICC, 0.68), and mean NPC distance (r = 0.43; ICC, 0.60) showing moderate test-retest reliability. ImPACT component scores demonstrated slightly stronger

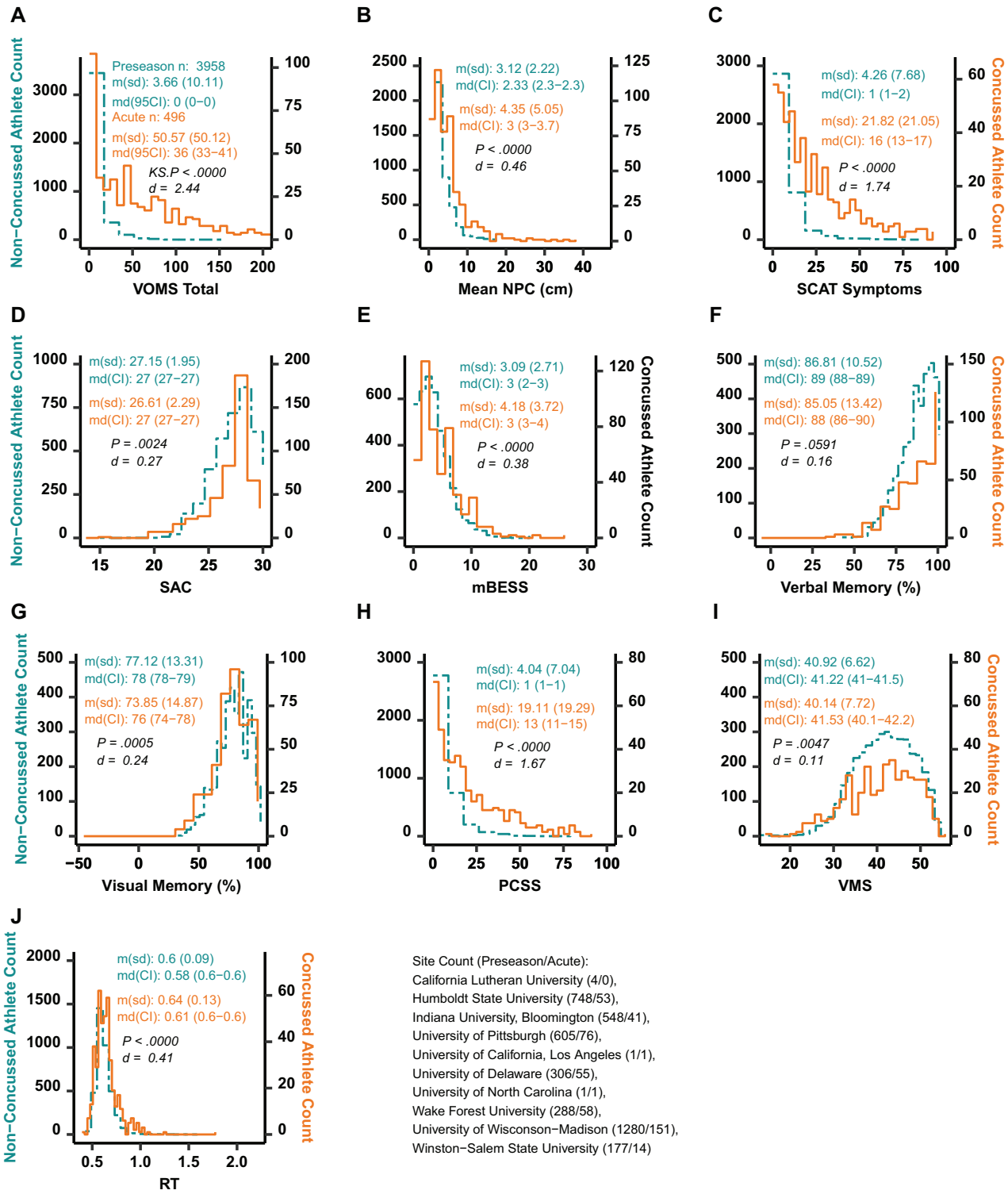

Figure 3. Distributions and descriptive statistics for Vestibular/Ocular-Motor Screening (VOMS), the third edition of the Sport Concussion Assessment Tool (SCAT3), and Immediate Post-concussion Assessment and Cognitive Testing (ImPACT) assessments. Preseason nonconcussed (dotted line) and acutely concussed (solid line) scores for (A) VOMS Total symptom severity, (B) mean near point of convergence (NPC), (C) SCAT3 Symptom Evaluation symptom severity (SCAT Symptoms), (D) Standardized Assessment of Concussion (SAC), modified Balance Error Scoring System (mBESS), (F) ImPACT Verbal Memory, (G) ImPACT Visual Memory, (H) ImPACT Post-Concussion Symptom Scale (PCSS) symptom severity, (I) ImPACT Visual Motor Speed (VMS), and (J) ImPACT Reaction Time (RT). The left axis displays the count of nonconcussed athletes, whereas the right axis displays the count of concussed athletes. Athlete evaluation counts for each site at the preseason and acute time frame are reported in the bottom right side of the figure. KS, Kolmorogov-Smirnov significance.

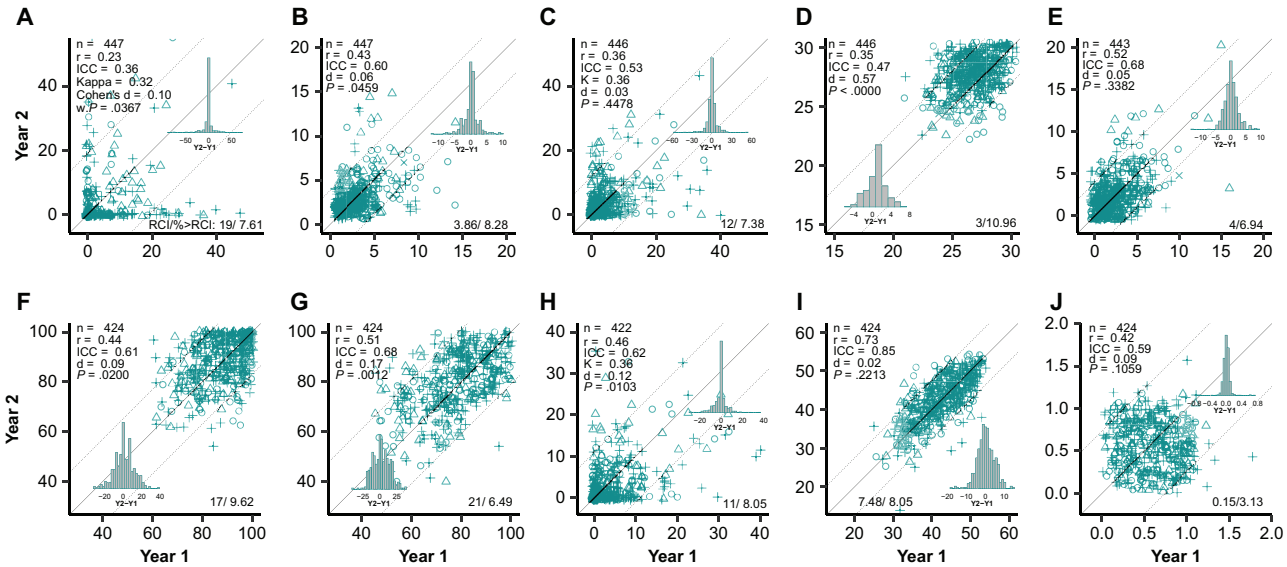

Figure 4. Test-retest reliability scatterplots for consecutive preseason evaluation scores on the Vestibular/Ocular-Motor Screening (VOMS), the third edition of the Sport Concussion Assessment Tool (SCAT3), and the Immediate Post-concussion Assessment and Cognitive Testing (ImPACT) assessments. Modified Brinley plots with mean difference histograms show the poor reliability of intrapatient change across preseasons for A) VOMS Total, B) mean nPC, C) SCAT3 Symptom Evaluation D) SAC, E) mBESS, F) ImPACT Verbal Memory (%), G) ImPact Visual Memory (%), H) ImPACT PCSS, I) Impact VMS, and J) Impact RT. Dotted lines indicate the boundary of the reliable change range, representing the +1 and –1 Reliable Change Index (RCI), respectively. The RCI for each tool and the percentage of athletes with performance changes outside the RCI band (% >RCI) are noted in the bottom right corner of each plot. Symbols indicate the sites of data collection: ×, California Lutheran University (n = 1), Δ, Humboldt State University (n = 91), ○, University of Pittsburgh (n = 128), +, University of Wisconsin-Madison (n = 227). ICC, intraclass correlation coefficient; mBESS, modified Balance Error Scoring System; PCSS, Post-Concussion Symptom Scale; RT, Reaction Time; SAC, Standardized Assessment of Concussion; VMS, Visual Motor Speed; w.p, Wilxocon significance.

correlations, with VMS (r = 0.73; ICC, 0.85) demonstrating the most consistency in performance between consecutive years, whereas RT (r = 0.42; ICC, 0.59) demonstrated the least consistency. Approximately 3% to 10% of athletes within the test-retest cohort recorded scores outside the RCI band (% >RCI) across testing years for all assessments, indicating a clinically meaningful change in test performance despite the lack of any known concussive injury. ImPACT RT (% >RCI = 3.13%) and Visual Memory (% >RCI = 6.49%) components had the least between-athlete variation outside of the expected range (eg, were most reliable), whereas ImPACT Verbal Memory (% >RCI = 9.62%) and SAC (% >RCI = 10.96%) demonstrated the most between-athlete variation. In the middle were VOMS Total and the SCAT3 Symptom Evaluation, with 7.61% and 7.38% of scores falling outside the RCI, respectively.

Subanalyses of sex-specific data (Appendix Figures A1-A6, available online) and LD/ADHD-specific data (Appendix Figures A10-A23, available online) revealed similar test-retest reliability results to the full data set. Female participants had slightly higher RCIs for symptom-based assessments (eg, VOMS Total, SCAT3 Symptom Evaluation, ImPACT PCSS) than did male participants. The data for athletes with a history of LD/ADHD were similar to the female athlete data, with higher symptom-based RCIs. The subset of athletes with a history of LD/ADHD also had a larger percentage of participants who recorded scores outside the RCI band, indicating slightly more variability across consecutive-year evaluations. There were not enough retest data points to enable test-retest analyses for the population of athletes with a history of migraine or the population with ≥3 previous concussions.

To reflect a more controlled testing environment that may be seen in smaller, traditional clinical settings, we repeated descriptive statistics and test-retest analyses after enacting 90%, 95%, and 99% CI outlier reductions. Most assessment tools displayed consistent maximum scores and consistent 90% CI RCI values across all CIs %. Most assessment tools displayed consistent median scores and 90% CI RCI values across all CIs % (Appendix Table A1, available online). However, mean scores varied when data beyond the 90%, 95%, and 99% CIs were removed. RCI values varied substantially across intervals, changing as much as 8 points for VOMS Total between the 90% (5 points) and 99% CIs (13 points) (Appendix Figure A13 and Appendix Table A1, available online).

Diagnostic Utility of PS Change Versus Raw Scores

After RCIs were generated for each tool from the test-retest analyses, PS Change scores were calculated for each assessment component. From there, ROC analyses

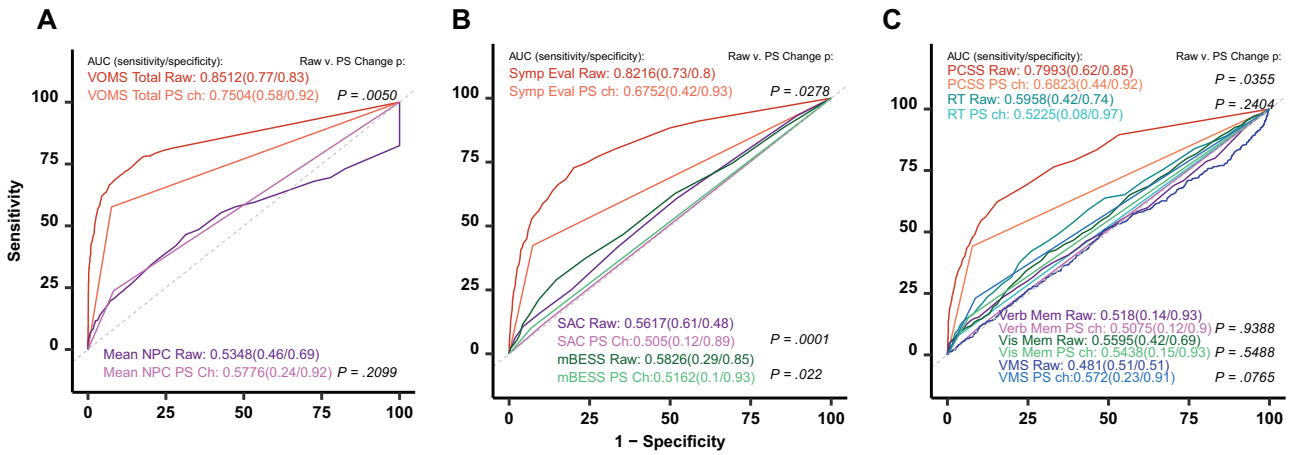

Figure 5. Receiver operating characteristic (ROC) curves for Vestibular/Ocular-Motor Screening (VOMS), third edition of the Sport Concussion Assessment Tool (SCAT3), and Immediate Post-concussion Assessment and Cognitive Testing (ImPACT) components. ROC curve analyses show the predictive utility between Raw and Preseason Change (PS ch) methods for (A) VOMS Total (red/orange) and mean near point of convergence (NPC) (purple/pink), (B) SCAT3 Symptom Evaluation (Symp Eval) (red/orange), Standardized Assessment of Concussion (SAC) (purple/pink), and modified Balance Error Scoring System (mBESS) (green/light green) scores, and (C) ImPACT Post-Concussion Symptom Scale (PCSS) (red/orange), Reaction Time (RT) (teal/cyan), Verbal Memory (Verb Mem) (purple/pink), Visual Memory (Vis Mem) (green/light green), and Visual Motor Speed (VMS) (blue/light blue). AUC, area under the curve.

were conducted, and the Youden method was used to determine the AUC, maximum sensitivity, and specificity for each assessment tool.[4] Raw and PS Change diagnostic utility for each assessment tool are graphed in Figure 5. The VOMS Total score (AUC = 0.8512; sensitivity, 77%; specificity, 83%) and the SCAT3 Symptom Evaluation (AUC = 0.8216; sensitivity, 73%; specificity, 80%) yielded the highest and second-highest predictive utilities for accurately identifying concussed individuals. The ImPACT PCSS also demonstrated higher levels of predictive utility, with an AUC of 0.80. All other tools and components displayed poor levels of diagnostic utility for identifying concussion, with AUCs ranging from 0.48 (VMS) to 0.60 (RT). No significant improvement in AUC accuracy was seen when preseason performance was incorporated into any tool (Delong method for AUC comparison; $P > .01$). In fact, VOMS Total showed a statistically significant decrease in AUC ($P < .01$) when PS Change scores were used.

Subanalyses of sex-specific data (Appendix Figures A1-A6, available online) revealed differences in performance between male- and female-only data. Raw score assessments of data for female participants demonstrated slightly higher AUCs than did assessments of data for male participants. For either sex, use of PS Change scores did not improve diagnostic utility for any assessment ($P > .01$). Athletes with a history of LD/ADHD, migraine, or ≥3 previous concussions were unable to undergo ROC analyses due to a lack of concussion observations.

ROC analyses were repeated using data within the 90%, 95%, and 99% CIs to examine whether removal of outlier data points affected diagnostic utility. Whereas the diagnostic accuracy for all tools improved with narrower CIs, there was no significant improvement in accuracy between using a same-day total versus a PS Change score for any tool under any of the CIs tested ($P > .01$) (Appendix Table A2, available online).

DISCUSSION

Preseason concussion baseline tests are commonly considered part of the standard in sports concussion management protocols.[14] However, the contribution of these performance scores to detect acute concussion continues to be questioned. This study is the first to use ROC analysis to directly compare the predictive utility between raw postinjury scores and individualized preseason baseline change score methods for the VOMS, SCAT3, and ImPACT tools in the same large population of collegiate athletes. The findings are stark: incorporating preseason scores did not meaningfully improve concussion identification. In the case of VOMS Total and SCAT3 Symptom Evaluation, incorporating preseason scores decreased predictive accuracy, indicating that clinicians' time may not be effectively used when conducting these resource-heavy assessments during standard preparticipation physicals.

The high levels of within-person variance seen in these tools directly contributed to the lack of meaningful improvement in diagnostic utility when PS Change scores were considered. This variance also led to poor to moderate retest reliability metrics regardless of the analytic method used (eg, Pearson r, ICC, κ). One possible explanation for this variance is the common design of these tools. All rely on subjective reporting from athletes, giving rise to possible data quality concerns from misinterpreted instructions

or responses, non–concussion specific responses, or potential underlying motivating factors behind responses.[2,43,45] Although these subjective tools have demonstrated a strong ability to differentiate between concussed and nonconcussed populations, there remains a need for qualitative assessment tools for concussion that could better detect subtle functional differences between concussed and nonconcussed individuals, such as video-based eye or pupil trackers, blood or fluid biomarkers, or reaction-time tests.[9,35,43,52]

It is possible that the CARE data set contains higher than normal levels of variance given the scope and size of the study. However, these levels of disparity have been documented previously in healthy populations. The SCAT3 Symptom Evaluation severity total score (ICC, 0.43-0.62[7,11]; κ = 0.41[7]), 5-word SAC (ICC, 0.29-0.39),[7,11,44] and mBESS (ICC, 0.54)[11] have consistently displayed poor to moderate levels of reliability. Studies on ImPACT have been more mixed, with Verbal Memory (ICC, 0.50), Visual Memory (ICC, 0.58), and RT (ICC, 0.47) demonstrating moderate reliability, whereas VMS (ICC, 0.72) has shown stronger consecutive-year relationships.[7] Even VOMS has demonstrated mixed results. VOMS has been shown to have high test-retest for short time frames.[55,56] Yet VOMS items have displayed significant variance across consecutive years, with κ correlations ranging from 0.28 (vertical saccades) to 0.51 (mean NPC distance).[7] Although not evaluated in this investigation, the effect of variation within the individual components of any subjective concussion assessment tool could be a topic of future research. Most concussion assessment tools, including VOMS, SCAT3/SCAT5, and ImPACT, contain multiple independent components. When assessing the combined tool's overall variance, the individual variances of these components must be summed.[31] In the case of VOMS, SCAT3, and ImPACT, the results from this analysis and others suggest that the combined variances of these tools are even larger, raising further questions regarding the use of preseason evaluation performance consideration with these tools in acute concussion evaluations.

These levels of variance directly affect change score calculations because both SD and Pearson r are directly used to calculate RCI. As such, it is not surprising to see how relatively low test-retest metrics relate to the relatively wide ranges of RCI values seen in the literature for SCAT3/SCAT5 and ImPACT. The similarly high test-retest variance seen in this study directly led to higher RCI values (Figure 4; SCAT3 Symptom Evaluation RCI, 12; VOMS Total RCI, 19). These high RCIs then directly affected the diagnostic utility of the tools by allowing a larger change in performance between preseason and acute SRC to be considered "normal." The higher the between-test variance, the larger the RCI and the larger change in performance an athlete must demonstrate to be accurately predicted as concussed or nonconcussed. Finally, it should be emphasized that RCIs operate under the assumption that a test has a normal distribution and varies smoothly about its mean. The results above show that the majority of VOMS, SCAT, and ImPACT components are nonparametric and heteroscedastic, challenging the applicability of any RCI information derived from these tools. Along with the fluctuating preseason performances of individual athletes observed in this and other studies,[7,11,44,55,56] these aspects should generate serious questions about the clinical utility of using RCI-derived individualized change scores from preseason testing sessions when assessing acute SRC.

Although the variance seen in these assessments may call into question their overall reliability for acute SRC identification, components of all tools showed fair to high discriminability between concussed and nonconcussed populations (eg, VOMS Total d = 2.45) and stable group-level performance across test time points. Individual performance may have changed from one year to the next, but group-level performance remained relatively stable, as shown by the minimal Cohen d values for each tool. This stability also showed the lack of any group-level learning effects—there was no systematic shift of the poor-performing or well-performing athletes. Each year some athletes perform well, whereas others do not.

Together, these discriminability and group stability factors directly led to the clinically useful levels of sensitivity and specificity seen in the ROC results. The overall utility for VOMS Total (AUC = 0.85), SCAT3 Symptom Evaluation (AUC = 0.82), and ImPACT PCSS (AUC = 0.80) was in line with current literature[11,38,41,44] and supported the continued use of these components in diagnostic batteries for concussion. Whereas these tools were beneficial for distinguishing between concussed and nonconcussed individuals, ROC analysis found no meaningful diagnostic accuracy improvement for using change from baseline for any tool or its components. Instead, VOMS Total, SCAT3 Symptom Evaluation, and ImPACT PCSS revealed significant reductions in predictive accuracy for identifying concussion when preseason scores were incorporated. Together, these findings emphasize how tools with high internal variance may receive no meaningful increase in accuracy when preseason performance is incorporated into diagnostic decisions.

It is possible that outlier data, sex, or LD/ADHD differences affected these analyses and reduced the test-retest metrics, RCIs, and/or AUCs for these tools. Whereas neurocognitive performance scores are known be vulnerable to confounding factors such as sex, the testing environment, sleep, premorbid medical conditions such as ADHD, academic performance, mental health status, or unresolved previous head injury,[12,34,37,48,51] there are currently no separate normative ranges or RCI for sex or LD/ADHD for ImPACT.[24] However, analyzing the data separately for sex and LD/ADHD groups and removing data beyond the 90%, 95%, and 99% CIs did not drastically alter either the test-retest or ROC results, indicating there is a level of inherent within-patient variance within VOMS, SCAT3, and ImPACT not driven by data quality or unique population characteristics alone. Conversely, outlier data removal did affect RCI calculations because of the use of the SD calculation. Moderate RCI shifts for VOMS Total and, to a lesser extent, the SCAT3 Symptom Evaluation and Visual Memory were seen and varied by as much as 11 between the 90%, 95% and 99% CIs, respectively.

Clinicians should keep these possible outlier effects in mind when considering individualized RCI performance changes as part of clinical diagnostic decisions.

The low performance of these tools as it relates to test-retest variance and preseason scores has been documented before. In their 2018 CARE test-retest analyses, Broglio and colleagues[7] noted common concussion assessments to possess "less than ideal" stability across time but acknowledged that preseason evaluation settings can be valuable for concussion education and for communication of an athletic organization's concussion protocols to young athletes and their families. In many organizations, preseason concussion education and evaluations have become a requirement for sports participation. Beginning in 2010, all NCAA member collegiate athletic programs were required to implement SRC education programs,[39] and all NCAA schools are required to conduct at least 1 preseason baseline concussion assessment for each student athlete.[27] Aside from collegiate programs, all 50 US states require some form of SRC education at the youth level.[42]

There certainly is merit to requiring preparticipation evaluations. General preseason physical examinations are important for any physical sport, as they can help identify athletes with preexisting conditions (eg, migraine history, motion sensitivity, concussion histories) that could influence postinjury concussion assessments. Whereas general preseason physical evaluations are important, this study's results echo those of others and indicate no improvement in the detection of acute SRC by including baseline concussion assessment testing as part of those evaluations. Not only was there no improvement in overall diagnostic accuracy, but the overall sensitivity of each tool decreased, some as much as 20%, with the incorporation of preseason baseline performance. This decrease suggests that clinicians may receive misdirecting information when preseason scores are incorporated. Physicians and organizations may want to consider separating baseline concussion assessments from general preseason physical evaluations and general education sessions to reduce resource strains while still ensuring that standards of care are met.

Limitations

This study was affected by many of the same limitations reported in previous SRC studies. This VOMS CARE Consortium analysis was based mostly (~77%) on athletes from NCAA Division I athletic programs and may not be representative of all athletes (eg, Division III, club sports, or other age ranges). As with all multicenter studies, CARE Consortium sites received comprehensive training on CARE protocols and test administration. Nevertheless, data were collected across multiple sites and years, introducing variability among sites and testers. Another limitation affecting many SRC studies is the motivation of student athletes at baseline testing in comparison to postinjury.[2,45] A lack of motivation by collegiate athletes at baseline could increase the test-retest variability and thus contribute to baseline test performance data's not being additive to acute concussion identification. Another issue possibly contributing to these results was the difference in testing environments between baseline testing and postinjury testing. Baseline testing is often conducted in small groups, whereas postinjury examinations are usually individualized. In both cases, the in-house medical staff (usually certified athletic trainers and/or team physicians but not neuropsychologists or psychometrists) conducts the examinations. Additionally, the number of previous baseline assessments, and possible testing fatigue during a lengthy preseason evaluation can affect the reliability of testing results. Finally, many components of the tools assessed are known to show significant skew and lack normality. Correlation methods such as ICC and reliability metrics such as RCI operate on the underlying expectation of a Gaussian distribution. Although the use of parametric-based statistics relying on mean and SD may not best reflect the data, it was warranted in this case because it allowed the current analysis to be directly compared with existing literature.

CONCLUSION

The test-retest reliability results for SCAT3, VOMS, and ImPACT from this study were consistent with the literature. The results from this study reinforced existing research and highlighted the large effect sizes and clinically useful diagnostic utility of VOMS, SCAT3 and ImPACT PCSS for identifying acute concussion. However, in this head-to-head comparison of VOMS, SCAT3, and ImPACT, we found no statistically significant increase in diagnostic accuracy for acute SRC when preseason baseline assessments were incorporated into acute evaluation performance. As demonstrated in this analysis, this reduction in utility was due to the poor to moderate test-retest reliability for all assessments. Health care providers and organizations may want to apply these findings to inform their cost-benefit analysis when deciding whether to use baseline SRC evaluations with their athletes.

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